

Data Analytics

Academic Year 2025-26

Course Project N.19: Autonomous Systems

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For this assignment you will work in pairs to carry out simple tasks of data analysis given a specific dataset. The goal of this assignment is to use Python and complementary libraries on a given dataset in order to *explore* and *analyze* the given data and *draw conclusions*.

Description

You are given a dataset with computers and a number of connected servers. Your goal is to analyze the graph and identify communities within it. Your tasks are to:

- Explore and describe the data (i.e., visualize the graph, compute centrality measures (degree centrality, closeness centrality, betweenness centrality), etc.). Which are the nodes with the highest influence? Any interesting observation?)
- Use two algorithms to identify communities and plot them, assigning a different color to each community.
- Identify the largest clique in the graph and visualize it accordingly, using a different color for its nodes.
- What is the minimum degree for a node to be considered a “heavy hitter”? Which nodes are heavy hitters?

Submission procedure and evaluation

You should produce a report of your work and its evaluation along with the source code. It will be a concise explanation of how you tackled the different tasks, the reasons of your choices, successive conclusions, graphs you produced, results of the decisions and their accuracy, *etc.*

Use Jupyter Notebook to produce results of the commands in a single .ipynb file. For more information check: <https://jupyter.org/documentation>

The report (max 5 pages) and the code of the project need to be submitted via iCorsi.

Please, upload all the required items in a single file and name it following the structure: **no_Project.[zip|tar.gz|7z]**. For instance, 05_projectname.tar.gz